

Total No. of Questions : 10]

SEAT No. :

P1358

[Total No. of Pages : 4

[4858] - 1106

T.E. (Information Technology)

SYSTEMS PROGRAMMING

(2012 Pattern) (Semester - II) (End-Sem.)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Your answers will be valued as whole.*
- 5) *Assume suitable data if necessary.*

UNIT - I, II & III

Q1) a) Explain how forward reference is handled in Single Pass assembler and solve it for the given example : **[6]**

```
START 202
MOVER AREG, = 5
MOVEM AREG , A
LOOP  MOVER AREG, A
      MOVER CREG, B
      ADD CREG, = '1'
      MOVEM CREG, B
      SUB CREG,A
      BC ANY, NEXT
      LTORG
      ADD CREG,B'
      BC LE LOOP
```

P.T.O.

```

NEXT    SUB AREG, = '1'
        BC LT, BACK
        STOP
        ORIGIN 219
        MULT CREG, B
A        DS 1
BACK    EQU LOOP
B        DS 1
        END

```

- b) Explain different parameter passing mechanisms in Macro-processor. [4]

OR

- Q2)** a) Compare how the four basic tasks of loader are performed in BSS loading scheme and DLL scheme. [4]
- b) Explain various advanced macro facilities like AIF, AGO, LCL, GBL, REPT and IRP statements with syntax and example. [6]

UNIT - I, II & III

- Q3)** For the code given below, show macro name table, macro definition table, Expanded code and the stack frame. [10]

```

MACRO
    XYZ &A, &B, &C
    READ &A
    ADD&B, ='5'
    PRINT & C
MEND
MACRO
    MIT &Z
    MACRO
        &z&W
        SUB &W, ='6'
        XYZ AREA, BREG, CR
        ADD &W, ='5'
    MEND

```

```

      ADD CREG , = '5'
MEND
START 200
MIT HELLO
MULT BREG , = '4'
HELLO BREG
PRINT C
C DS 1
END

```

OR

- Q4)** a) List down various phases of compiler. What are functions of Lexical Analyzer? Explain patterns, tokens and lexemes with examples. [6]
 b) Give the flowchart for pass I of DLL scheme. [4]

UNIT - IV

- Q5)** a) Differentiate between top down parser and bottom up parser. [4]
 b) Generate SLR parsing table for the grammar given below and parse the string $id1 + id2 + id3 * id4$ [10]

Grammar:

$E \rightarrow E + T / T$

$T \rightarrow T * F / F$

$F \rightarrow id$

- c) Explain operator precedence parser. [4]

OR

- Q6)** a) Design LR(1) parser for the given grammar. Also show the moves by the parser for input string "ab". [10]

$S \rightarrow aABb$

$A \rightarrow c/\epsilon$

$B \rightarrow d/\epsilon$

- b) Compare SLR and LALR parsing methods. [4]
 c) Define Handle and handle pruning w.r.t. bottom up parser. For the grammar given, [4]

$S \rightarrow 0S1 / 01$

Identify the handles at each step and parse the string 000111.

UNIT - V

- Q7)** a) Explain the following : [8]
- i) Dependency graph of Type expression.
 - ii) Synthesized and inherited attributes.
- b) Write the context free grammar for following and also perform syntax directed translation for the same into three address code. [8]

while condition do S

OR

- Q8)** a) Explain the need of Intermediate code generation in compiler. Generate quadruple and triple for the given expression: $a = b + c * d$. [8]
- b) Write the method of generating intermediate code for the Boolean expression. [8]

UNIT - VI

- Q9)** a) Obtain the TAC for the following code before and after applying the optimization techniques using. [12]
- i) Removal of Loop Invariants
 - ii) Elimination of common sub expressions

for ($i = 1$; $i \leq 10$; $i++$)

$X[i][2*j-1] = Y[i][2*j-1]$

- b) Explain different intermediate code generation techniques. [4]

OR

- Q10)** a) Discuss code generation issues. [4]
- b) Discuss with suitable example machine dependent code optimization. [8]
- c) Write a short note on activation record. [4]

